Project 2: Coronary Arterial Network Modeling

Nicolas Garcia Callejas

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During the making of this Project, we were required to treat this network as a set of circuits with different resistances / stenosis percentages. I will say that trying to do this project was challenging as I was trying to keep the project was organized and friendly for change in parameters as possible. However, the implementation of data classes into this application serves quite useful for writing my parameters, paths, and variables needed in the project, but the learning curve on having to place a int, float or str for each parameters was tedious and honestly a bit too much for my liking, while it is beneficiary if you know the data type being used for input and what should be expected for output. Nevertheless, the overall goal of this project was to develop a coronary arterial network and test different bypass options based on the .csv file provided. Before getting into the specifics of how each bypass performed and provided the best flow for the overall system we must set the bar, and mention that the "not occluded" model had a total outlet flow of 9.023e-03 mmHg*cm^3/s. This as the base of our comparison we can make more sense of our comparison. When it came to determining the best graft option, we analyzed this by comparing the original branch's flow to a restored flow. The restored flow of the branch was calculated by the difference between the total flow and the baseline flow. In our given cases we concluded based on the restored outlet flow (ranked better to worst) Graft 1 restored the most outlet flow (1.902e-04 mmHg*cm^3/s). Secondly, Graft 3 had a restored outlet flow of 1.719e04 mmHg*cm^3/s, again slightly below Graft 1, and lastly Graft 2 had a restored outlet flow of 1.539e-04 mmHg*cm^3/s. In this assignment we were also asked to investigate % of stenosis or vessel occlusion, changing it from the baseline 0.01, to 0.10, 0.30, 0.60. To avoid redundance with numbers, we noticed that by increasing the % of stenosis in the system the less outlet flow was restored for all of the grafts provided, the most noticeable change was after increasing past 30%. Another parameter that we played around with was with increasing or decreasing the systemic pressure, by doing this we saw that as we increase the pressure to 1000mmHg the restored outlet flow decrease and the total outlet flow decreased and vice versa when the pressure decreased.

In summary this project served as a great starting point for understanding how to use the data classes library and how tedious it can be to know and implement the data type of each added variable, in addition to learning to implementing more cell data types into our vtk file for visualization. While some improvements can be made such as adding an animation into our project and seen the pressure drop over time, there are some limitations on what we can achieve while learning libraries in python such as the paraview library that might include this type of functions.

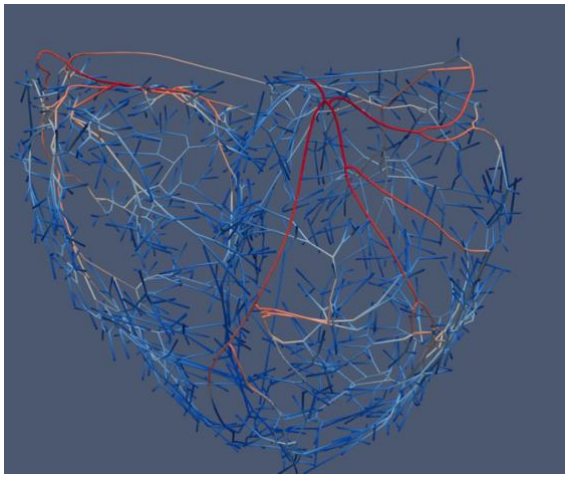

Figure 1: Coronary Solution, No occlusion, No graft

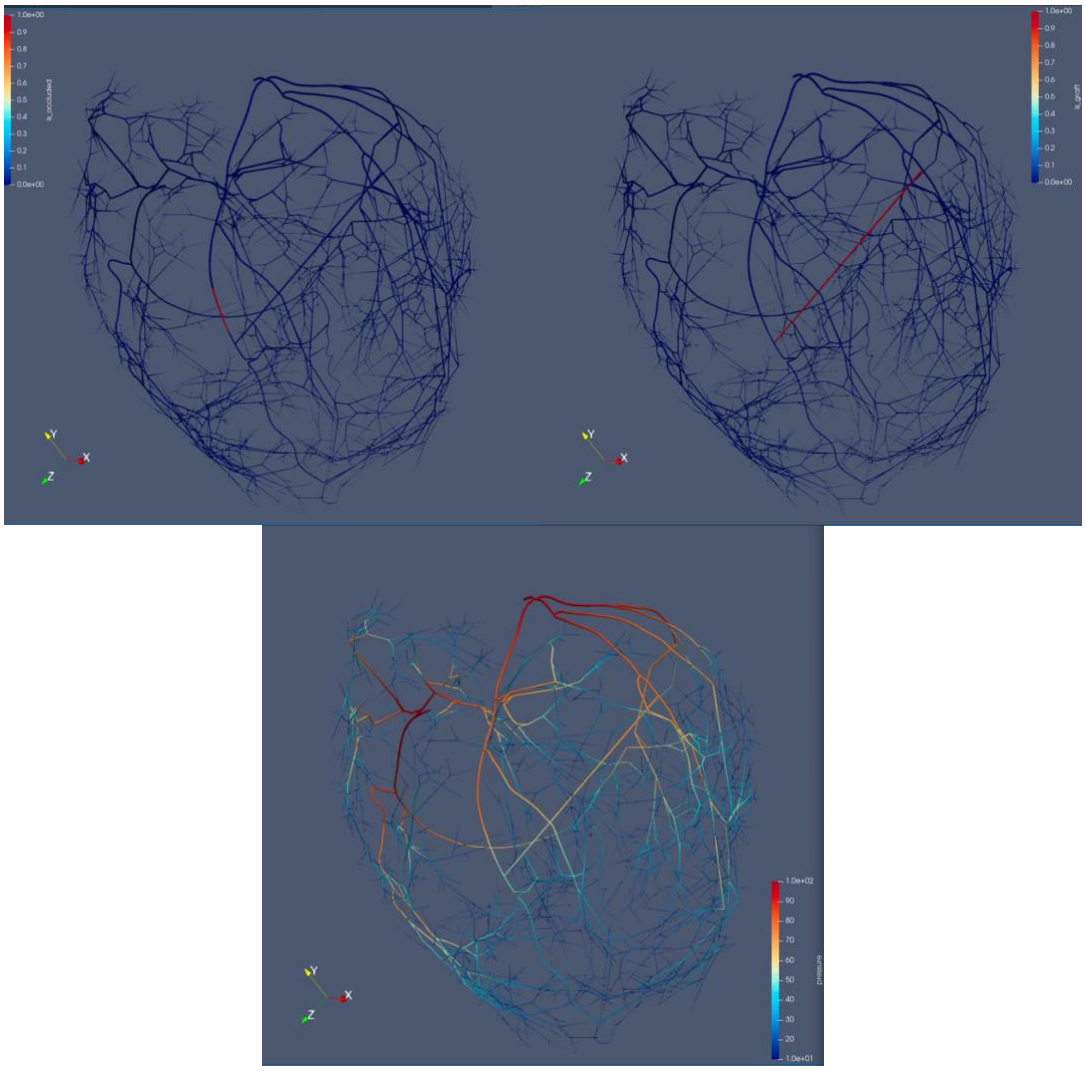

Figure 2: CA Solution, Occlusion(top left) with graft 0 (top right)

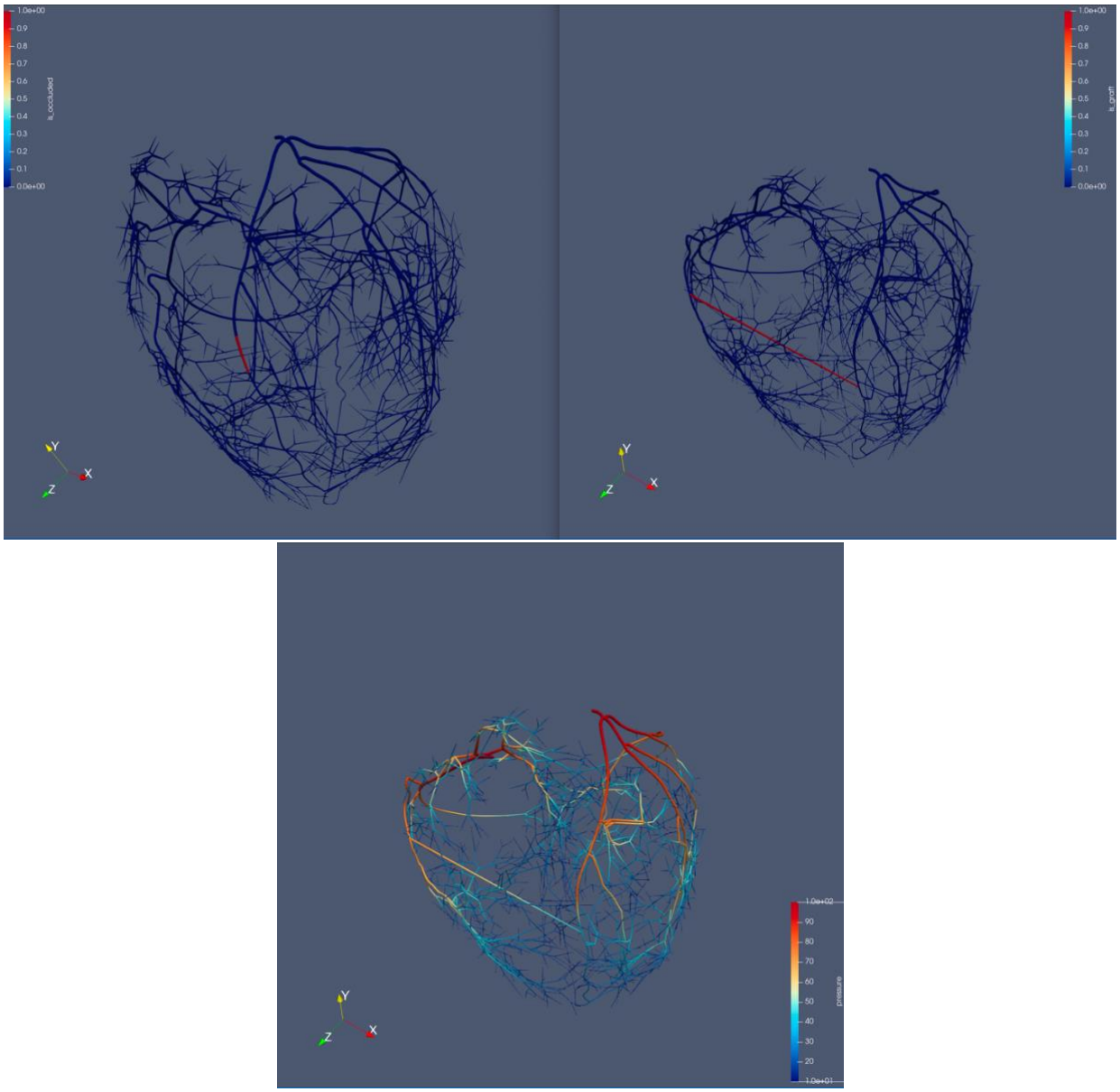

Figure 3: CA Solution, Occlusion(top left) with graft 1 (top right)

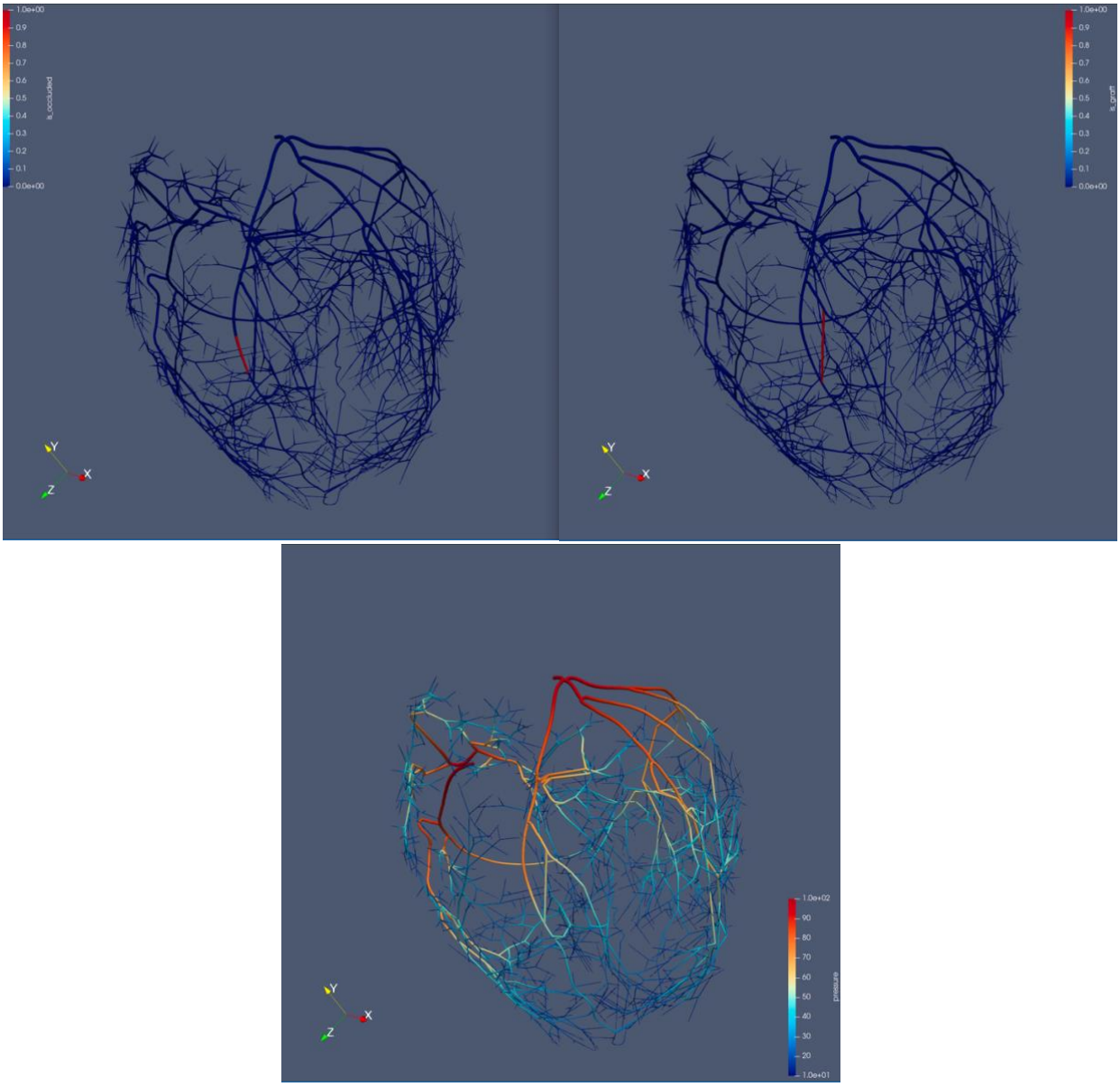

Figure 4: CA Solution, Occlusion(top left) with graft 2 (top right)